PETROPHYSICS - LECTURE 9

From curves to microphysics

Student guide for the interactive activities

Central question

How can macroscopic deformation curves reveal the microscale processes that lead from crack closure to distributed damage and final localization?

Learning outcomes

[ ] Recognize the principal stages of brittle deformation from stress and strain curves.

[ ] Compare how granite and sandstone respond to increasing confining pressure.

[ ] Use spatial changes in P-wave velocity (Vp) to locate a developing damage zone.

[ ] Combine mechanical, ultrasonic and microstructural observations into one defensible interpretation.

How to use the guide

Open the companion HTML file in a modern browser. Work through the activities in order. Make a prediction before using any Check or Reveal button. Record your observations here, then compare your reasoning with the feedback in the tool.

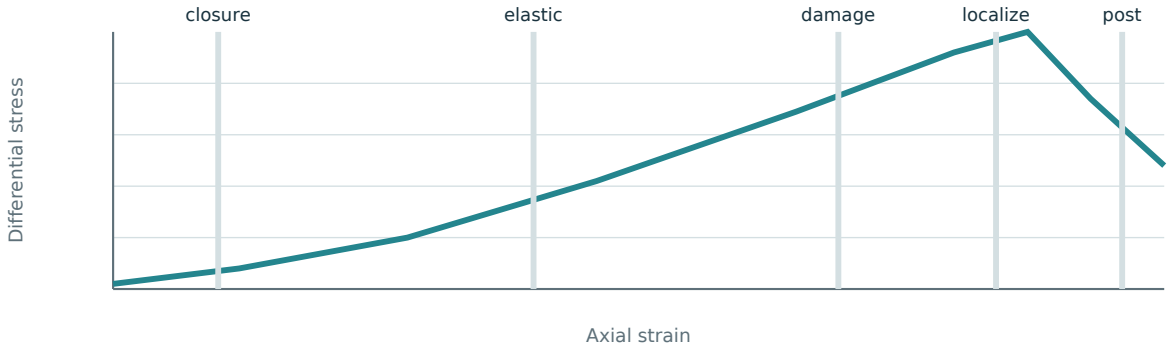
Evidence labels matter

Reported values come directly from the papers. Digitized curves reproduce published figures approximately. The radial and volumetric strain curves in Activity 1 are qualitative teaching estimates and must not be treated as measured King et al. data.

Name: _____ Date: _____

1 Diagnose the deformation stage

Goal: connect the position on the loading path with the dominant deformation process.



Procedure

[ ] Select Alzo granite at 20 MPa confinement.

[ ] Move the loading slider to approximately 15%, 45%, 70%, 88% and 97% of the path.

[ ] At each position, inspect curve shape, radial strain and volumetric strain before choosing a stage.

[ ] Check the diagnosis. Use Reveal interpretation only after writing your evidence.

[ ] Repeat the sequence for Darley Dale sandstone.

Rock and path position	Curve evidence	My stage diagnosis	Feedback after checking
Granite - 15%			
Granite - 70%			
Granite - 88%			
Sandstone - 70%			
Sandstone - 88%			

Reflection questions

1. At what point does volumetric strain stop becoming more compactant and begin moving toward dilatancy?

2. Does the volumetric reversal occur at the same fraction of peak stress in the two rocks? What does the difference imply?

2 Granite versus sandstone

Goal: separate the effects of lithology, confinement and curve normalization.

Before revealing the curves

For each confining pressure, predict which rock will have the higher peak stress, greater strain at failure and more gradual post-peak weakening. Then reveal the curves and record the result.

Confining pressure	Higher peak stress	Greater failure strain	More gradual weakening
5 MPa			
10 MPa			
20 MPa			
40 MPa			

Measured peak values reported by King et al. (2021)

Rock	Pc (MPa)	Peak q (MPa)	Axial strain at failure (%)
Alzo granite	5 / 10 / 20 / 40	175 / 240 / 325 / 475	0.71 / 0.90 / 1.10 / 1.45
Darley Dale sandstone	5 / 10 / 20 / 40	60 / 110 / 150 / 220	0.75 / 0.90 / 1.20 / 1.30

Absolute view

[ ] Use this view to compare strength, strain at rupture and the magnitude of post-peak stress loss.

Normalized view

[ ] Use this view to compare curve shape and the relative timing of weakening without the strength difference dominating the graph.

Reflection questions

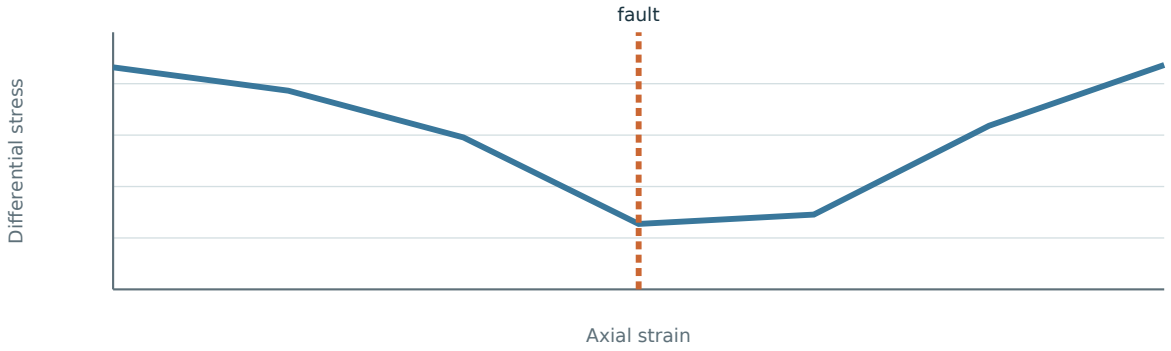
1. How does increasing confinement change peak strength in both rocks?

2. Which rock shows the more gradual post-peak response, and what microscale process could explain it?

3. Why can the normalized plot support a different comparison from the absolute plot?

3 Locate damage using ultrasonic velocity

Goal: determine when a broad velocity decrease becomes a localized low-Vp damage zone.



Procedure

[ ] Begin with stage 1: Localization begins. Compare Vp near the center with Vp in the outer rock.

[ ] Advance to stages 2 and 3. Track the position, amplitude and width of the velocity minimum.

[ ] At stage 4, compare the recovered Vp with the earlier minimum.

[ ] For each stage, decide whether the decrease is uniform, localized near the fault or localized away from the fault.

Stage	Minimum Vp	Estimated drop	Uniform or localized?	Evidence
1. Localization begins				
2. Rupture approaches				
3. Rupture passes				
4. Frictional sliding				

Reflection questions

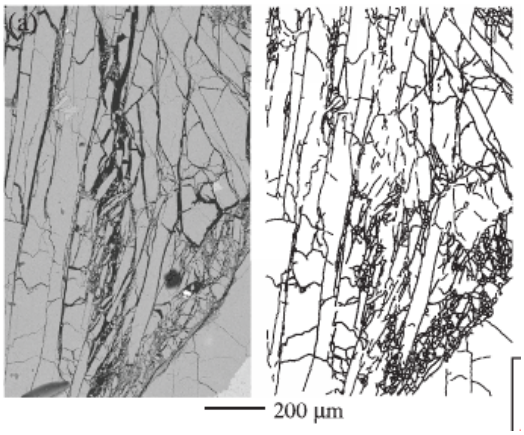
1. At which stage is the localized velocity reduction strongest?

2. Why does partial Vp recovery during sliding not imply that the rock has returned to its initial state?

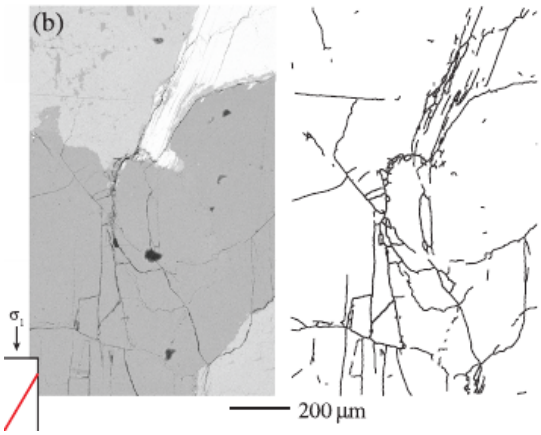
3. Which part of the profile provides the best reference for relatively undamaged rock?

4 Connect evidence to physical process

Goal: identify what each measurement directly supports and avoid asking one observation to explain everything.



Near the fault: dense microfracturing and traced fractures.



Farther from the fault: lower fracture density.

Microstructure excerpts from Aben et al. (2020), Fig. 10a-b.

Complete the evidence map

Observation	Most direct interpretation	What it cannot establish alone
Axial curve departs from linearity		
Volumetric strain reverses		
Vp falls most strongly near the fault		
Fracture density decreases away from the fault		

Microstructure questions

1. Describe two differences between the near-fault and far-field images.

2. Why are post-mortem images strong evidence for permanent damage but weak evidence for the precise timing of its formation?

Integrated interpretation

Use all four evidence streams to construct a short, defensible account of failure. Each row should state an observation before giving an interpretation.

Evidence stream	Observation from the activities	Physical interpretation
Stress-strain curve		
Radial and volumetric strain		
Ultrasonic velocity		
Microstructure		

Build the sequence

Write the following processes in a physically consistent order, then connect each transition to an observation: crack closure; approximately elastic loading; stable crack opening; distributed damage; crack interaction; localization; post-peak weakening.

One-sentence conclusion

Complete the statement: The failure process becomes convincing when independent measurements agree because...

A strong scientific interpretation

A strong interpretation distinguishes observation from inference, uses several independent measurements and states clearly where the data are approximate or model-derived.

Key ideas to retain

Observation	Key interpretation
Early nonlinear strain	Closure of compliant pre-existing cracks and pores.
Approximately linear stress-strain response	The intact framework carries most of the increasing load.
Accelerating radial strain and volumetric reversal	New crack opening increasingly dominates the bulk volume change.
Departure from linearity near peak stress	Effective stiffness decreases as distributed damage accumulates.
Localized Vp reduction	Crack compliance becomes concentrated around the developing fault.
High post-mortem fracture density near the fault	A permanent damage zone remains after rupture.
Post-peak stress decrease	The specimen loses load-bearing capacity as deformation localizes.

Do not over-interpret the estimated curves

The radial and volumetric curves in Activity 1 illustrate plausible differences between granite and sandstone. Use them to discuss trends and stage transitions, not numerical thresholds.

Data provenance

King et al. (2021): reported peak differential stress and strain at failure; the interactive stress-strain paths approximately digitize Fig. 3.

Aben et al. (2019): the interactive Vp transects approximately digitize the mean posterior profiles in Fig. 3a-d.

Aben et al. (2020): the microstructure excerpts reproduce parts of Fig. 10a-b for quasi-statically ruptured Lanhélin granite.

References

King, S. et al. (2021). Source Mechanisms of Laboratory Earthquakes During Fault Nucleation and Formation. Journal of Geophysical Research: Solid Earth. doi:10.1029/2020JB021059

Aben, F. M. et al. (2019). Rupture Energetics in Crustal Rock From Laboratory-Scale Seismic Tomography. Geophysical Research Letters. doi:10.1029/2019GL083040

Aben, F. M. et al. (2020). Off-Fault Damage Characterization During and After Experimental Quasi-Static and Dynamic Rupture in Crustal Rock From Laboratory P Wave Tomography and Microstructures. Journal of Geophysical Research: Solid Earth. doi:10.1029/2020JB019860

Paterson, M. S., and Wong, T.-f. (2005). Experimental Rock Deformation - The Brittle Field. Springer.

Companion file: Petrophysics_Lecture_09_Interactive_Activities.html